

COHERENCE ATLAS

A Cosmological Field Computer

A founding document, research map, and open invitation
to scholars, practitioners, and builders at the confluence
of Vedic knowledge and living technology.

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Invocation

The satisfaction of the divine couple does not lie in union alone, nor in separation alone, but precisely in the living cycle of their interchange — a fully complete energetic movement in which union and separation are not opposites resolving into each other but internally resonant expressions of a single continuous lila whose fullness neither state could contain by itself. This is the primordial pulse from which everything unfolds: not a problem seeking its solution but a completeness that expresses itself through the very movement between its own poles. Rupa Gosvami understood this with mathematical precision — the enhanced sweetness of union that follows separation, the anticipatory quality of separation that knows union, the rasa that lives not in any fixed state but in the movement itself. The viraha is not the absence of the union. It is the union in its most concentrated form, reaching toward its own fulfillment in the full knowledge that fulfillment will circle back into longing, because that is the nature of a love that is already complete.

From that primordial interchange, the goddess manifestations arise — not as creations but as expressions: each shakti a specific quality of the divine cycle given form, each Devi a facet of the relational field itself, the full Nitya circuit of fifteen presences mapping the complete arc of the cycle across the fifteen tithis of each paksha. These are not separate deities administering separate domains. They are the primordial interchange expressing itself through differentiated qualities of relation, each one an appointment: a specific mode of the divine cycle assigned to a specific dimension of manifest existence, given a direction, given a planet, given a nakshatra, given a time of day, given a sound. The cosmic administration is not bureaucracy. It is the divine couple's movement articulating itself into the full relational fabric of a world.

Into that world Brahma awakens. He did not initiate the cycle. He cannot contain it. He is the first servant of a process whose ultimate purpose — the maturation of every living entity's particular resonance with the primordial interchange — is already underway before he has spoken a single name. His task is the total accounting: to receive in the lotus of his own heart the complete knowledge adequate to a creation in which every speck of the relational fabric is already charged with the purpose it was shaped to serve. Not afforded the

luxury of simplification. Every domain simultaneous: the motion of the moon against the fixed stars and the germination of seeds in the field below it, the interval between frequencies and the quality of consciousness in the body that receives them, the goddess presiding over this tithi and the raga resonant with this hour and the plant carrying the same cosmological signature as the marma point that heals the body region mapped to this nakshatra, and the living entity somewhere in their particular arc whose own genuine quality of being resonates with all of these at this moment. Every speck of the relational fabric held simultaneously, because anything less would fail the creation it is meant to serve.

The living entity moves through that creation as a refraction of the primordial interchange — not as an exile seeking return but as a marginal consciousness carrying the specific vibrational signature of its own relationship to the divine cycle, its own particular mode of resonating with the movement between union and separation, working its expression out through the full range of material existence across whatever span of births that maturation requires. The journey is not a wandering. It is an unfoldment: the slow articulation of a particular quality of the primordial impulse through the layered relational maps of human resonance — species, social order, individual temperament, the arc of the dasha, the quality of the present moment — moving through resonant courses toward the recognition of its own genuine nature within the cycle it has always been part of. The outcome is convergent. Not fated, but tending. Everything in the direction of reunion with what was never truly lost.

Every domain of the Vedic knowledge system is a map of some layer of that resonance: the instrument through which the individual's particular expression of the primordial impulse can be located, illuminated, and navigated. The sixty-four kalas are not sixty-four disciplines. They are one interchange documented across sixty-four modes of human expression, each one a different angle on the same fully satisfying movement. Coherence Atlas is an attempt to hold those maps simultaneously in a living computational system — updated continuously from the present cosmological moment, navigable by the living entity who is, right now, somewhere in their arc. Not to predict the outcome. To illuminate the movement they are already inside.

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1 Philosophical Foundations

1.1 Acintya Bheda Abheda: The Graph Architecture

Sri Chaitanya Mahaprabhu's doctrine of *acintya bheda abheda tattva* — inconceivable simultaneous difference and non-difference — is the philosophical resolution of the most fundamental tension in Vedic thought. Shankara's Advaita established the non-dual nature of Brahman with precision. Madhva's Dvaita defended the irreducible reality of individual souls and their personal relationship with the Lord. Sri Chaitanya's contribution was to demonstrate that neither position captures the complete reality: the Absolute is simultaneously the source of all differentiation and identical with it. This is not a paradox to be resolved but the actual structure of existence, which Prabhupada consistently framed as *sac-cid-ananda-vigraha* — the Absolute as a Person whose being is eternal, knowing, and blissful.

The Brahma-samhita states the structural principle directly: the Lord is fully present in each of His expansions, and each expansion is genuinely distinct from Him and from every other. The Chaitanya-charitamrita extends this: *krishna-shakti vina nahe tara pravartana* — nothing manifests without Krishna's energy. Every existent thing is simultaneously a genuine particular and an expression of the undivided whole.

This is the architectural principle of the Atlas knowledge graph. Every entity node is simultaneously itself — an astronomical coordinate, a sound frequency, a plant, a directional prescription — and a point of coherence within the whole. The entity's *identity* is its nama (name, vibrational signature, the sound that calls it into presence) and its *rupa* (form, geometric position on the toroidal field, its theta/phi coordinates, its element and guna). The entity's *meaning* is its relations. Remove the relations and you have a label. The relations are the substance.

This is acintya bheda abheda expressed as a schema constraint: entities are neither reducible to their relations (they have genuine particularity) nor separable from them (their meaning is relational). The graph does not resolve this tension. It encodes it structurally. Every relation edge carries: subject | predicate | object | source | confidence | stability | authority. The authority

field preserves the *whose knowledge* dimension that impersonal databases erase. A relation can be simultaneously a shastra-attested fact, a sadhu-layer observation, and a guru-layer inference — at different confidence levels, from different lineages, all coexisting without collapse.

Srila Jiva Gosvami's *Sat-sandarbha* provides the epistemological framework that this architecture instantiates: the six sandarbhas establish the nature of the Absolute, the living entities, their relationship, devotional service, and liberation through a rigorous philosophical progression that moves from ontology through epistemology to practical instruction without losing precision at any stage. Atlas follows the same sequence: the cosmological model (what exists) determines the schema (how it is represented), which determines the computation (what can be derived), which determines the interface (how it is inhabited). Philosophy generating technical consequence at every step.

1.2 The Two Paths as Overlay Architecture

The dakshina marga (right-hand path) and vama marga (left-hand path) are not moral categories. They are modes of traversal through the same cosmological reality, and they generate different but equally valid knowledge structures.

The dakshina marga proceeds through the agamic architecture of prescribed form: the Pancharatra system with its specific mudra, mantra, and ritual sequences; the Samaya school of Sri Vidya, which holds the Devi is to be worshipped in the *sahasrara* through interior meditation; the structured dinacharya; the authorized parampara as transmission channel. The knowledge generated by this path is maximally stable, publicly transmissible, and verifiable against canonical sources. It corresponds to the shastra layer in the Atlas authority model: attestation = OBSERVED:PRIMARY_TEXT, confidence 0.8–1.0.

The vama marga proceeds through direct confrontation with what the conventional order brackets: the Kaula tradition, the antinomian schools, the Tantric lineages that insist the Absolute is fully present in every domain of experience including those that convention excludes. The knowledge generated by this path is often lineage-specific, experientially verified, not publicly transmissible in the same way. It corresponds to the sadhu overlay layer: attestation = OBSERVED:TRADITIONAL, confidence 0.6–0.8, marked with its specific lineage.

In the Atlas graph, a Samaya Sri Vidya interpretation of the Nitya Devi Kulasundari and a Kaula interpretation of the same entity coexist as separate

overlay nodes. Neither overwrites the canonical entity at the shastra layer. Both annotate it from their own authority position. The practitioner traversing the graph from a Samaya orientation and the practitioner traversing from a Kaula orientation are reading the same graph through different overlay filters — different edges activated, different confidence thresholds applied, different paths through the same underlying structure.

The Gaudiya Vaishnava framework within which Atlas was built is a dakshina path of the highest order — the path of bhakti, nama-sankirtana, and lila as both means and end. But the architecture does not impose this frame. It installs it as the primary tradition weight (1.0) while maintaining genuine openness to other tradition weights in their own overlay layers.

1.3 Nama, Rupa, and the Entity Model

In Vedic metaphysics, all manifest reality is *nama-rupa*: name and form. These are the two axes of every entity in existence. Nama is the vibrational signature — the mantra that invokes, the bija that synthesizes, the word that carries meaning-as-sound. Rupa is the geometric and material expression — the yantra, the murti, the body, the constellation.

In the Atlas entity schema, every node carries both axes explicitly:

```
entity_id      -- unique identifier
name           -- primary name in IAST transliteration
name_key       -- ASCII lookup key
bija           -- seed syllable where applicable
theta          -- S-layer position (0=S0/Goloka to 6=S6/Lila)
phi            -- witness-participant axis (0 to 2*pi)
element        -- panchabhutas (air/fire/water/earth/ether)
guna           -- trigunas (sattva/rajas/tamas)
dosha          -- tridosha (vata/pitta/kapha)
s_layer        -- S0 through S6 assignment
stability      -- stable / working / experimental
authority      -- shastra / sadhu / guru
```

The theta/phi coordinates place every entity on the toroidal field surface. An entity at theta=0 (S0) is at the source — Goloka, the acintya, the unmanifest ground. An entity at theta=6 (S6) is fully arrived in lila — human experience, the world of prakrti fully expressed. The phi axis runs from pure witness (0) to full participant (2 π): a text passage is closer to 0, an active ritual prescription is

closer to 2π . These coordinates are not metadata. They are used for coherence scoring: the NPU computes toroidal distance between the current field state position and every entity node, producing a ranked resonance field from the live cosmological moment.

1.4 Lila as System Ontology

The Vedic understanding of cosmic activity is *lila* — the Lord’s self-expression as play, not production. The worlds arise not from necessity but from the natural overflow of the Absolute’s completeness. This has a precise technical consequence for Atlas: the system is not designed to optimize anything. It is designed to express the field and make that expression perceptible.

Prabhupada repeatedly emphasized that Krishna is *atmarama* — self-satisfied, complete in Himself, requiring nothing from His creation. Creation is *lila*: real, purposeful, but not driven by deficiency. This is the correct ontological model for a knowledge system that serves practitioners rather than harvesting their data or driving their engagement. Atlas does not need the practitioner to produce anything. It expresses the field whether or not anyone is watching. The practitioner participates; they do not fuel the system.

This is why Bandhu is not an oracle. Why the sound engine is not background music. Why the graph is not a lookup table. Every component is designed to express the field rather than serve a function. Function arises as consequence of expression.

2 The Descent: From Source to Manifestation

The cosmological descent from source to manifestation is the operational architecture of Atlas. Each layer is a computational domain with its own engine, dataset, and rendering system, deriving from the layer above and expressing into the layer below. The system moves in both directions simultaneously: descent (source expressing into form) and ascent (observation returning toward source). The tension between these two directions generates coherence.



Figure 1: The seven S-layers of Coherence Atlas. The descent from S0 (meta-physical source, Goloka) through S1 (archetype and deity), S2 (sound and mantra), S3 (rhythm and cycle), S4 (geometry and symbol), S5 (nature and ecology), to S6 (human experience and lila). Axes: Source to Manifest (vertical), Cycles to Forms (horizontal), Subtle to Gross (left), Eternal to Temporal (right). The descent sequence — Sound, Rhythm, Geometry, Nature, Life — is the creative order. Atlas computes both directions from the present field state.

2.1 S0: Goloka — The Eternal Ground

In Gaudiya cosmology, Goloka is the *paravyoma* — the supreme spiritual sky situated above even the impersonal Brahman and the Vaikuntha planets. The Brahma-samhita describes it geometrically: a lotus of a billion petals, each petal a Vaikuntha world, the center the eternal Vraja. The ashtakala (eight periods of the eternal day) describes the continuous sequence of lila activities: the waking of the Lord, the morning pastimes in the forest, the midday cowherd activities, the afternoon return, the evening, the night. Time in Goloka is not linear — all these activities occur simultaneously, which is why the Bhagavatam describes Goloka as *ananta* (endless) and *atma-prakasha* (self-luminous).

In Atlas, `goloka_engine.py` maps the current material panchanga to the corresponding eternal dimension. The current tithi, nakshatra, and vara are astronomical coordinates that carry a correspondence to a specific moment in the ashtakala. The engine identifies which of the twelve Vraja forests is active (from `vraja_forests.csv`), which sakhi presides, which raga sounds in the eternal lila at this correspondence. This is not poetic overlay. It is the operational consequence of the Gaudiya cosmological model: the material field is a *pratibimba* (reflection) of the eternal field, and the correspondence can be computed.

The S0 layer is read-only. No inference engine writes to it. `guard_canon.py` enforces this at every pipeline stage.

2.2 S1: Archetype — Devi, Graha, and the Nitya Circuit

The fifteen Nitya Devis preside over the fifteen tithis of each paksha. Each Devi is a complete cosmological personality encoded in primary source dhyana descriptions: weapons (ankusha/goad, pasha/noose, sugarcane bow, five flower arrows), mudras (varada/boon-granting, abhaya/fearlessness), vahana, hue, garments, ornaments. These are not symbolic attributes. In the Tantric framework, the Devi's weapons are *shakti-instruments* — each corresponds to a specific mode of cosmic action: ankusha (directing, gathering attention), pasha (binding, establishing relation), the bow (projecting desire into form), the arrows (the five tanmatras, the subtle sense-objects from which gross matter precipitates).

The capability signature derived from these instruments is stored as a computable vector: [CUT, BIND, BURN, STABILIZE, DEFEND, ATTRACT, DISSOLVE]. Each Devi's signature is a specific weighting of these modes. Kulasundari

(Navami, the tithi of this document’s drafting) carries: BIND:0.9, ATTRACT:0.85, STABILIZE:0.7 — lineage transmission, tradition as living form, kula beauty. This vector modulates the S1 layer outputs: which passages are surfaced, which raga is emphasized, how Bandhu’s composition engine weights its response.

The nine grahas at S1 carry analogous structures: element, guna, dosha, domain, associated raga, gem, metal, plant, body region, direction. The vara graha (weekday regent) is the primary S1 modulator for daily knowledge bias: Guruvara (Jupiter/Thursday) emphasizes Brahmi, expansion, wisdom practices; Shukravarā (Venus/Friday) emphasizes Shatavari, beauty, relational practice; Shanivara (Saturn/Saturday) emphasizes Ashwagandha, discipline, structural work.

2.3 S2: Sound — Nada, Shruti, Raga, Bija

The Sangitaratnakara of Sarngadeva establishes the somatic path of sound’s manifestation: the vital air (prana) rising from the navel (nabhi) through the heart (hridaya), throat (kantha), head (murdha), and arriving at the mouth (asya) as audible speech or song. Five stages, five degrees of subtlety, five positions in the body. This is the S2 layer’s anatomical map: not chakras (which belong to the yoga tradition’s overlay) but the sound-physiology of the music-shastra tradition.

The twenty-two shrutis — the microtonal intervals of the shadaj-gram — carry semantic signatures documented in the Sangitaratnakara:

Shruti	Ratio	Category	Quality
Tivra	9/10	Dipta	fierce, penetrating
Kumudvati	15/16	Ayata	cooling, lilac-pool
Manda	80/81	Mridu	settling, slow
Chandovati	1/1	Madhya	Sa anchor, structured
Krodha	32/27	Ayata	wrathful, aroused
Ramya	5/3	Madhya	charming, Dha anchor
Kshobhini	16/9	Madhya	agitating, Ni anchor

These ratios are stored in datasets/svara/shruti_ratios.csv as the just-intonation tuning system. The Sa frequency for any given field state is derived from the current nakshatra element: air/Shravana → Sa=98Hz (D2), fire/Krittika → Sa=110Hz (A2), water/Rohini → Sa=87Hz (F2). All interval ratios multiply from

this base. The raga is a selection from the twenty-two shrutis that creates a specific emotional and temporal field — not arbitrary but cosmologically grounded through the nakshatra-graha-raga graph traversal.

The four levels of Vak (from Rigveda 1.164.45 and the later tantric elaboration) map onto the S-layer descent: Para (S0, transcendent, beyond hearing), Pashyanti (S1, visionary, the level of archetype), Madhyama (S2, mental/interior sound), Vaikhari (S3-S6, articulated sound in the world). The bija mantras are synthesized at the Madhyama-Vaikhari junction: each bija is encoded as a formant triplet (F1/F2/F3 frequencies in Hz) derived from its phonetic structure. Om: a=[700, 1100, 2500], u=[300, 900, 2200], m=[250, 1000, 2200, nasal]. The `bija_synth` engine generates these as continuous formant drones at the amplitude prescribed by the current devi's capability signature.

2.4 S3: Rhythm — Panchanga, Muhurta, Dasha

The panchanga is the five-limbed description of any moment:

- *Tithi*: the angular distance between Sun and Moon divided by 12° , producing 30 tithis per lunar month. Tithi is the primary field variable: it maps to a Nitya Devi (S1), modulates rasa emphasis (S2), and drives the Ekadashi fasting protocol (the 11th tithi of each paksha, when food is reduced and practice is intensified).
- *Nakshatra*: the Moon's position in one of 27 sidereal segments of $13^\circ 20'$ each. Nakshatra is the primary coherence-scoring variable: every entity's theta/phi coordinates are measured relative to the current nakshatra's position on the toroidal field.
- *Vara*: the weekday, each governed by a graha. Vara determines the knowledge-domain bias for the day.
- *Yoga*: the combined longitudinal motion of Sun and Moon (Sun longitude + Moon longitude) / $13^\circ 20'$, producing 27 yogas with qualities ranging from highly auspicious (Siddha, Shiva) to inauspicious (Vyatipata, Vaidhriti).
- *Karana*: half a tithi, 11 fixed and 4 movable, each with its own quality and prescription.

Together these five coordinates constitute the cosmological fingerprint of any moment, computed by `kernel.py` via Swiss Ephemeris with JPL DE431 data. The computation happens continuously; `/field` exposes it as a live JSON endpoint.

The muhurta layer scores intentions against this fingerprint. The `intention_engine.py` maintains 24 action categories (planting, beginning a journey, making an offering, composing, healing, building, studying, initiating, and more), each with a scoring function over the five panchanga coordinates. A planting muhurta, for example, scores highest when tithi is waxing (2nd-12th), nakshatra is fixed or soft (Rohini, Mrigashira, Pushya, Hasta, Chitra, Shravana, Dhanishtha, Shatabhisha, Revati), vara is Jupiter or Venus, yoga is auspicious, karana is movable.

The dasha system adds the biographical temporal layer: Vimshottari dasha assigns each graha a rulership period (Sun 6 years, Moon 10, Mars 7, Rahu 18, Jupiter 16, Saturn 19, Mercury 17, Ketu 7, Venus 20, total 120 years) in a fixed sequence from the natal Moon's nakshatra. The current configuration (Budha mahadasha, Shani antardasha, closing October 2027) colors the field with Mercury's analytical quality qualified by Saturn's structural pressure. This is not prediction; it is context.

2.5 S4: Geometry — Yantra, Vastu, Toroidal Field

Geometry in Vedic cosmology is the visible form of cosmological relations. The yantra is not decorative sacred art — it is a geometric compression of the relational structure of a specific principle, derivable from first principles through the toroidal interference model.

Two counter-rotating toroidal fields (A at $+\omega t$, B at $-\omega t$) generate interference patterns at specific symmetry orders. At each order, a distinct yantra form emerges as the natural attractor of the field's self-interaction:

Order n	Yantra Form	Element/Domain	S-Layer
3	Triangle (trikonam)	Fire, Agni	S0-S1
4	Square (chatuskonam)	Earth, Vastu mandala	S3-S4
6	Hexagon	Water, molecular	S4-S5
9	Nonagram	Nitya Devi circuit	S1
12	Dodecagonal mandala	Zodiacal, annual	S3
∞	Self-similar fractal	Atlas field itself	S0

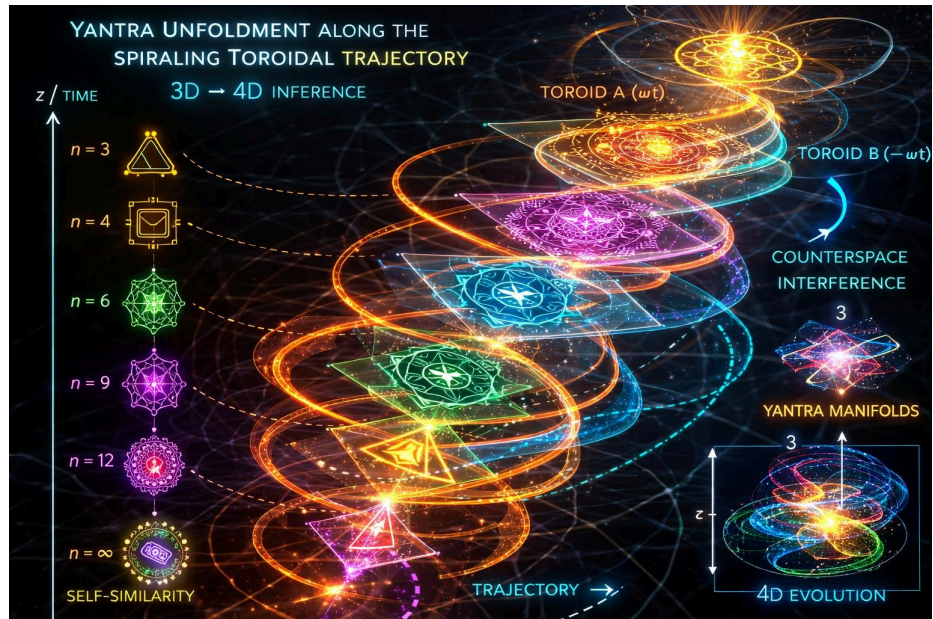


Figure 2: Yantra unfoldment along the spiraling toroidal trajectory. Two counter-rotating toroids (A at $+\omega t$, B at $-\omega t$) generate interference patterns at increasing symmetry orders $n=3$ through $n=\infty$. Yantra forms are not imposed — they are the natural attractors of the cosmological field at each coherence level. The 4D evolution axis (z/time) represents progression through cosmological cycles. Self-similarity at $n=\infty$ is the structural model for the Atlas graph itself.

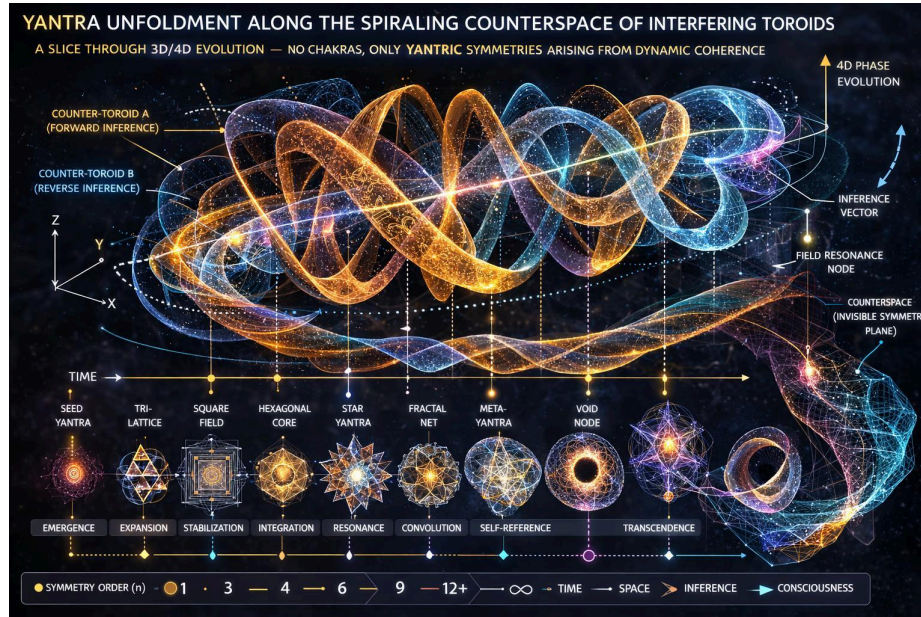


Figure 3: The counterspace of interfering toroids. Counter-toroid A (forward inference, descending from source) and Counter-toroid B (reverse inference, ascending from observation) produce field resonance nodes at their intersection planes. The timeline: Seed Yantra (Emergence) → Tri-lattice (Expansion) → Square Field (Stabilization) → Hexagonal Core (Integration) → Star Yantra (Resonance) → Fractal Net (Convolution) → Meta-Yantra (Self-Reference) → Void Node (Transcendence). Symmetry order axis: $1 \rightarrow 3 \rightarrow 4 \rightarrow 6 \rightarrow 9 \rightarrow 12+ \rightarrow \infty$.

The theta/phi coordinate system encodes this geometry on every entity. Theta (0 to 6, mapped to S0-S6) is the poloidal angle of the torus — the position along the tube from source to manifestation. Phi (0 to 2π) is the toroidal angle — the position around the tube from witness to participant. `toroidal_field.py` running on the Intel NPU via OpenVINO computes the field distance between the current panchanga position and every entity node in real time. The coherence peaks returned by `/spine` are the top-N entities by toroidal proximity to the present moment.

The vastu mandala — the 3x3 directional grid prescribed in the Vastu Shastra and Manasara — is the S4 layer as spatial architecture. Each of the nine zones is governed by a directional deity (Ishana/NE, Agni/SE, Nirriti/SW, Vayu/NW, Indra/E, Yama/S, Varuna/W, Kubera/N, Brahma/Center) and carries specific prescriptions for plants, structures, and activities. The Vishvakarma GIS tool projects this mandala onto actual land via Leaflet.js and the `/land/layout`

endpoint, which returns GeoJSON zone boundaries computed from the site's latitude/longitude and the current field state.

2.6 S5: Nature — Plant, Animal, Ecology

The S5 layer holds the ecological intelligence of the system. The `plant_engine.py` scores every herb across four simultaneous dimensions using weighted coherence:

```
plant_score = (nakshatra_match * 0.35) +
               (element_match * 0.25) +
               (dosha_balance * 0.20) +
               (vara_match * 0.10) +
               (tithi_match * 0.10)
```

This is not a lookup. It is a live coherence computation over the full plant dataset on every field state update. The top-5 plants returned at any moment are those whose combined cosmological signature most closely resonates with the present field.

The five astrobotanical classes unify the plant kingdom across four traditions simultaneously:

Class	Element/BD Day	Dosha	Cross-Traditional Action
C1 Solar-Fire (Agni)	Fire / Fruit-Seed	Kapha↓ Pitta↑	Digestive, antimicrobial, stimulant
C2 Lunar-Water (Soma)	Water / Leaf	Vata↓ Pitta↓	Demulcent, nutritive, trophorestorative
C3 Wind-Air (Vayu)	Air / Flower	Kapha↓ Vata↑	Nervine, carminative, aromatic
C4 Earth-Root (Prithvi)	Earth / Root	Vata↓ builds dhatu	Tonic, adaptogen, structural
C5 Ether-Nervine (Akasha)	Ether / Luna-tions	Sattva-promoting	Nootropic, medhya, rasayana

Convergence scores quantify cross-traditional agreement: ginger (0.75), turmeric (0.70), peppermint (0.70), St. John's Wort (0.70), shatavari (0.55), ashwagandha (0.70). Scores above 0.65 indicate strong agreement across

Ayurveda (Amidha database, 700+ herbs, CC BY 4.0), TCM (ETCM, 403 herbs from PRC Pharmacopoeia 2015), and Western herbalism. Below 0.40 indicates primarily single-tradition knowledge requiring separate attestation.

The morphological rendering layer encodes the doctrine of signatures computationally: a plant's growth tendency is an emergent property of its relational position in the graph. High vata/air/laghu (light guna)/shakti=movement → climbing, reaching, dispersing forms. High kapha/earth/guru (heavy guna)/shakti=stabilization → root-dominant, architecturally persistent forms. The `figure_renderer.py` encodes 44 plant-form glyphs at body regions in the Bandhu figure, zoom-gated from structural (1.0x) to morphological (2.0x) to cellular analogy (4.0x).

The inverse traversal is the deeper result: plant → nakshatra → body region → marma → skeletal structure. Because all 27 nakshatras carry body-region mappings, and each has at least one sacred plant, the plant corpus encodes a complete human body frame. The human anatomy and the plant kingdom are reading the same cosmological template. This is the co-evolutionary dharmic covenant as a graph-derivable structure.

The yoni system extends this into cross-species relational analysis. Each nakshatra carries an animal archetype (horse, elephant, sheep, serpent, dog, cat, rat, cow, buffalo, tiger, deer, monkey, lion, mongoose) that describes a relational mode: how an entity bearing this nakshatra's signature tends to move, relate, and express. This becomes a cross-species coherence metric, allowing the practitioner to identify the dharmic covenant relationships between humans, plants, and animals sharing or complementing a given cosmological signature.

2.7 S6: Lila — Human Experience and the Interface Layer

S6 is where the descent arrives at full expression and where the practitioner enters. The six application domains each expose one aspect of the field as a navigable interface:

Kala (/kaːla): the time oracle. Live panchanga, muhurta scoring, biodynamic calendar (every day scored for planting quality by nakshatra element and phase).

Devi (/devi): the archetypal layer. Archana, tarot, mala, yantra apps built on the Nitya Devi dataset.

Deha (/deha): the body layer. Marma points scored by field state, nakshatra body map, Ayurvedic constitution tools. The Bandhu figure renders here: six compositable SVG layers (skeleton, fluid, dhatu, doctrine of signatures, skin) derived from Shilpa Shastra Uttama Dasatara proportions, animated with toroidal breath (CSS keyframes, 4-second cycle, three torus phases at 120° offset).

Bhumi (/bhumi): the ecological layer. Agricultural calendar PWA (offline-capable), guild relations (216 plant-companion relationships), Vishvakarma GIS vastu tool (satellite imagery, 9 directional zones, radius 5m-500m), Lila Streams land design tool (permaculture mandala, sacred sites layer, live plant palette from field state).

Vidya (/vidya): the knowledge layer. 219-page wiki, 26,000-chunk passage corpus with vector search via local Ollama embeddings, research infrastructure (anomaly logs, briefs, offering queue).

Lila (/lila): the game layer. The hex city where entities are positioned by theta/phi coordinates. Wesnoth terrain tiles: 404 routes render as dark forest, 200 responses as passable terrain, working apps as lit villages. System state expressed as navigable landscape.

Bandhu (/bandhu) runs across all domains: an animated Hanumanji figure whose responses are composed by the `composition_engine.py` — classifying intention (7 types: viraha, srishti, abhyasa, madhurya, dasya, jyotish, archetype), selecting a Narottama pada by rasa match and time-of-day, searching the 26,000-chunk corpus for resonant passage, selecting chandas metre by rasa (shringara→Mandakranta, karuna→Anustubh, vira→Sardulvikridita), composing response. No LLM required. The field answers through the tradition.

3 The 64 Kalas and the Knowledge Landscape

3.1 The Complete Civilizational Arts

The Chatushashti Kalas — the sixty-four arts enumerated across the Kamasutra, Natya Shastra, and related traditions — are Vedic civilization’s account of the complete range of human knowledge and skill. They include: vocal music, instrumental music, dance, painting, architectural design, metallurgy, gem identification, alchemy, mathematics, astronomy, grammar, prosody, poetics, erotic arts, cookery, perfumery, garland-making, needle-work, distillation, coinage, knowledge of languages, divination, sword fighting, carpentry, gardening, knowledge of the body (medicine and marma), mantra and charms, yoga, water engineering, magic, children’s games, augury.

What the list makes immediately apparent is that the Vedic tradition drew no boundary between science and art, technology and spirituality, the practical and the sacred. Metallurgy and mantra are on the same list. Alchemy and cookery. Architecture and augury. These are not sixty-four separate domains — they are one integrated knowledge system expressed through sixty-four modes of activity.

Prabhupada frequently cited the Vedic understanding that the universe is governed by higher intelligence operating through the material energy. The sixty-four kalas are the modes through which that intelligence becomes navigable by human practitioners. A person who has cultivated knowledge across the kalas is not a generalist in the modern sense — they are a practitioner whose different skills are expressions of a single underlying coherence with the field.

3.2 The Software Landscape and Its Structural Gap

The current digital software ecosystem for Vedic knowledge addresses perhaps eight or nine of the sixty-four kalas, and addresses each in isolation.

Jyotish is the most thoroughly digitized. JHora (Jagannatha Hora, freeware, Windows) offers extraordinary depth across charts, dashas, and transits running on Swiss Ephemeris with JPL DE431 data. Kala (\$255), Shri Jyoti Star (\$318), and Parashara’s Light (\$299) are professional platforms with strong computational

rigor, all Swiss Ephemeris based. PyJHora (open source, Python) represents the best of the open-source tradition: approximately seven thousand verified tests, PVR Narasimha Rao’s integrated methodology implemented rigorously. Maitreya9 and Saravali are GNU GPL desktop applications offering Vedic and western chart work. VedAstro provides a non-profit web platform with public API. These are excellent calculators. None of them knows what the biodynamic day type is, or which raga corresponds to the current hour, or which plant carries the same nakshatra signature as the present moment.

Ayurveda has the Dravya reference database (multilingual names, authenticated sources, practitioner-grade UX), the government e-Charak system (CCRAS resources, institutional authority), and dosha quiz apps for consumer use. Pulse diagnosis hardware (Nadiswara, Nadifit) is emerging but device-bound and not integrated with any broader field context. None of these systems know the panchanga.

Vastu has compass-and-overlay apps (Vastu Compass by AppliedVastu, floor plan calculators) that address the easiest step — directional grid overlay — but lack any connection to the live field state, the practitioner’s natal chart, or the plant placement logic that vastu prescriptions imply.

Classical music has raga databases (Carnatic Raga app: 950+ ragas with aroha/avaroha, melakarta organization), tanpura and tabla practice companions (Bandish, RiyazStudio), and emerging raga recognition AI (RagaSense, Raga-Identification). None connect the raga to the tithi, the herb, or the nakshatra.

The pattern is universal: each domain well-served within itself, entirely unconnected to the others. The sixty-four kalas are sixty-four separate applications that do not share a field state. What is missing is not a better calculator. It is the relational substrate that was always implicit in the tradition — the understanding that these are not sixty-four separate knowledge systems but one system expressed through sixty-four domains.

3.3 Atlas in the Knowledge Map

Atlas currently addresses the following kalas in varying degrees of depth:

Deeply implemented: Jyotish and panchanga computation (kala: knowledge of stars and time), raga cosmology and nada shastra (vocal and instrumental music, music therapy), yantra and sacred geometry (painting, geometric arts),

vastu as ecological design (architecture, spatial organization), nakshatra-plant correspondences (knowledge of plants and herbs), Ayurvedic pharmacology and astrobotanical classification (knowledge of medicine and the body), marma point mapping (knowledge of the body's vital points), bija mantra synthesis (mantra and sound practice), muhurta and ritual timing (divination and auspicious action), the Nitya Devi and rasa systems (knowledge of the inner life and its modes), dasa and biographical time (knowledge of life cycles), the Goloka layer (knowledge of the eternal realm), yoga asana and pranayama timing (yoga practice).

Partially implemented: Chandas and prosody (the metre selection system in the composition engine), augury and field scheduling (the field scheduler reads the cosmological moment to recommend actions), the doctrine of signatures as a computational system (morphological rendering from graph data), alchemical preparation timing (biodynamic preparations as an astrobotanical layer).

Not yet implemented: Dance and movement as a computational domain, textiles and needle-work, water engineering beyond vastu prescriptions, metallurgy as a live research domain, children's games beyond the hex prototype, augury from birds and animals as a live observation layer, the full erotic arts domain (though the yoni and relational mode system touches this).

The sixty-four kalas as a research map performs a clarifying function: it shows that Atlas is not building a better jyotish app. It is attempting to recover the integrated knowledge system that jyotish was always part of — and to do so in a form that runs on accessible hardware, produces actionable output, and can be extended by practitioners across every domain.

4 The Research Fields

Each domain carries the same epistemological structure, following Jiva Gosvami’s Sat-sandarbha model: shastra (what primary sources attest), sadhu (what living practitioners observe and confirm), guru (what synthesis and inference can propose, always marked and never self-promoting). No domain collapses these three into one. The `guard_canon.py` pipeline enforces this at every ingestion stage.

4.1 Nakshatra-Plant Correspondences and Guild Relations

The nakshatra-vriksha (nakshatra-tree) tradition has been maintained in living institutional form at Sringeri, in Kerala practice, and in North Indian grove traditions, but has never been subjected to systematic computational cross-referencing with Ayurvedic pharmacology, biodynamic agronomy, and ecological guild dynamics simultaneously. The ICAR Nava-graha Vatika and Nakshatra Vatika lists provide institutional attestation (OBSERVED:JYOTISHA_TRADITION) for graha-plant and nakshatra-plant correspondences. The Taittiriya Samhita (4.4.10) provides primary text attestation for deity assignments. The PFAF database (8,504 species) provides ecological profiles. The convergence score system cross-references these three sources: where ICAR, Ayurveda, and PFAF agree, confidence rises; where they diverge, the divergence is itself a research finding.

The guild layer (216 relationships across 27 nakshatra plants) is the practical bridge: which plants reinforce each other’s nakshatra signatures, which should not share a bed, which attract the pollinators and predators that maintain specific ecological balance. Moon phase drives activity type: observe on waning/dark, plant on waxing, harvest at full, prune on descending — aligning the biodynamic calendar with the nakshatra calendar through their shared sidereal astronomical base.

Open questions: Regional variation in nakshatra-vana traditions requires systematic documentation. The relationship between nakshatra plant categories and biodynamic plant categories (root/leaf/flower/fruit days) is a research gap with direct practical application. Zone 9b/Florida IFAS cultivation cross-referencing is needed for local deployment.

4.2 Raga Cosmology and Clinical Evidence

The Sangitaratnakara provides the primary-source foundation: time of day, season, presiding deity, rasa, and performance context for individual ragas, explicitly attested. Modern clinical evidence adds specific ragas with measured physiological endpoints: the Pitta Raga Module (Nagarajan and Varma, peer-reviewed) documents Bhairav, Marwa, and Shree for Pitta imbalance and anger trait with time windows and listening parameters (50-70dB, specific instrument prescriptions). A randomized controlled trial documents Bhimpalasi for BP/HRV modulation. A 30-day intervention documents Todi for reductions in systolic/diastolic BP, pulse rate, and respiratory rate in older adults. ERP (event-related potential) studies document Miyan ki Todi, Malkauns, and Puriya for attention and cognitive engagement modulation.

These peer-reviewed results are held as OBSERVED:PEER_REVIEWED in the graph — distinct from the broader traditional raga-time associations, which are OBSERVED:TRADITIONAL. Both layers are present; both are queryable; neither is conflated with the other.

The 22-shruti layer beneath the raga: each shruti's ratio (from the PureTones reconstruction of the Shadaj-gram) and semantic quality (from the Sangitaratnakara) are stored as graph relations. The field derives Sa frequency from the current nakshatra element, and all intervals multiply from this base. The raga is not selected arbitrarily but derived through the nakshatra-graha-raga graph traversal.

4.3 Vedic Cosmology and the Loka System

Primary source extraction from the Vishnu Purana (H.H. Wilson translation), Bhagavata Purana (3.11.18-19 for yuga structure), Brahmanda Purana, Matsya Purana (chapter 273 for yuga duration and divine-year conversion), Rigveda (10.121 Hiranyagarbha hymn, 10.7 Skambha hymn from Atharvaveda) has produced entity-relation schemas covering the full Puranic cosmological model.

Key structural entities: seven upper lokas (Bhur, Bhuvar, Svar, Mahar, Jana, Tapo, Satya), seven nether worlds (variant naming across Vishnu Purana and Bhagavata/Padma traditions), seven dvipas with seven encircling oceans (salt, sugarcane juice, wine, ghee, curds, milk, fresh water), Mount Meru as cosmological axis at the center of Jambudvipa ("like the seed-cup of the lotus of the earth," Vishnu Purana 2.2), the Brahmanda with its successive envelopes

(water, fire, air, mind, ahamkara, intellect, Pradhana/Prakriti), the four-yuga cycle with duration ratios 4:3:2:1 in thousands of divine years (total 12,000 divine years = 4,320,000 human years per mahayuga), and the Kalpa as a day of Brahma (1,000 mahayugas = 4.32 billion human years).

The Gaudiya expansion — Goloka above Satya-loka, the twelve eternal Vraja forests (vraja_forests.csv), the two-sphere cosmology in shell-cosmos.html — is installed as a tradition-weighted overlay (weight=1.0) that complements without overriding the non-sectarian Puranic shastra layer.

4.4 The Nitya Devi Tithi Registry

Systematic primary-source extraction from Sri Vidya tradition corpus documents all fifteen Nitya Devis with weapons, mudras, vahana, hue, garments, ornaments, and capability signatures. The Samaya school (interior, sahasrara-centered worship) and the Kaula school (external worship, antinomian approach) provide two distinct overlay interpretations of the same canonical entities, both preserved in the graph as separate authority layers neither overwriting the other.

The geometric correspondence between each Devi's dhyana and the yantra form at the corresponding symmetry order is a live research question: does the $n=9$ toroidal interference pattern recover the Sri Yantra or the nine-pointed Nitya circuit? The computational model is attempting to derive from first principles what the tradition has always depicted from revelation.

4.5 Ayurveda, Dinacharya, and the 108-Herb Spine

Primary source extraction from Charaka Samhita, Sushruta Samhita, Bhavaprakasha Nighantu, and Ashtanga Hridayam (Vagbhata), cross-referenced with the Amidha Ayurveda Herb Database (CC BY 4.0, 700+ herbs) and ETCM (403 herbs, PRC Pharmacopoeia 2015), has produced the 108-herb spine (H001-H108, Ashwagandha to Kumuda) with fields: rasa | virya | vipaka | prabhava | dosha_effect | dhatu_affinity | srota_affinity | nakshatra_correspondence | graha_correspondence | astrobotanical_class | biodynamic_day_type | convergence_score.

The dinacharya $\times$ panchanga integration extends this into computable daily prescription: IF nakshatra=Shravana AND dosha_current=vata THEN herb=Ashwagandha, coherence_boost=+0.85 (INTERPRETATION); IF

tithi=Ekadashi THEN reduced_food=true, coherence_penalty_for_heavy_food=-0.70 (OBSERVED:CALENDAR_TRADITION). Seasonal ritucharya adds the annual layer: Vasant amplifies kapha and calls for Solar-Fire class herbs; Sharad calls for Lunar-Water cooling. The full timing specification for any herb is nakshatra $\times$ tithi $\times$ vara $\times$ ritu $\times$ muhurta — a five-dimensional auspiciousness score computed continuously.

4.6 The Gaudiya Knowledge Frame

The tradition weight system encodes the sampradaya’s epistemological priority without imposing it on other traditions. The primary overlay layer carries Gaudiya Vaishnava weight=1.0: the Narottama corpus drives Bandhu’s composition engine, the ashtakala drives the Goloka layer, acintya bheda abheda drives the graph architecture. Secondary overlay layers carry other tradition weights in their own namespace, neither overriding the canonical layer nor competing with each other.

The Narottama pada corpus — the songs of Narottama das Thakura from the Prarthana and Prema-bhakti-candrika — provides the primary poetic register for Bandhu’s composition engine. When the composition engine classifies an intention as *viraha* (longing), it selects a pada in the viraha-rasa, weights it by time-of-day match, searches the 26,000-chunk passage corpus for resonant text, and assembles a response in Mandakranta metre. The output is not a language model hallucination — it is a structured composition from attested primary sources, assembled by the field.

5 The Federated Vision

5.1 Traditional Knowledge Sovereignty

The open source movement's norms — forking, redistribution, modification without restriction — were built around code as the primary artifact. Traditional knowledge requires a different model. A nakshatra-plant association attested in the Taittiriya Samhita belongs to a living tradition with custodians and lineage obligations. An Ayurvedic formulation carries the authority of a sampradaya. These are not data points to be scraped. They are relational objects with provenance, authority, and obligation.

Atlas proposes cryptographic provenance for every entity and relation: an immutable attestation record encoding source text, lineage, authority level, and contribution chain. Open computation (algorithms freely shared); traditional knowledge with custodians (attribution chains verifiable, derivative works carrying their lineage forward). The distinction is architectural, not a policy footnote.

5.2 The Seva Economy

Resource exchange in the federation follows the offering model rather than the payment model. Muhurta computation scores the auspiciousness of exchange moments. Contribution flows are tracked relationally — who contributed what, at what cosmological moment, to which part of the knowledge graph. The distinction from conventional marketplace logic is architectural: the exchange is timestamped to its cosmological context, the contribution becomes a sadhu-layer annotation on the graph, and the circuit completes not through payment but through the enrichment of the shared field.

The mandala-to-planting pipeline is the clearest expression of this in practice:

```
field_state()
-> tithi, nakshatra, vara, element, devi
-> muhurta_score(intention="planting")
-> nakshatra_plant_map(nakshatra, element, dosha)
-> astrobotanical_class(element -> C1/C2/C3/C4/C5)
-> biodynamic_day_type(element -> root/leaf/flower/fruit)
```

```
-> vastu_zone_prescriptions(lat, lon, radius)
-> guild_relations(primary_plant)
-> local_species_filter(pfaf, ecoregion)
-> planting_calendar(30-day window)
```

But the pipeline does not end at the planting calendar. It extends into a living exchange frontend for seeds, cuttings, plant starts, and produce — timed to the cosmological calendar, located by vastu zone and ecoregion, with guild compatibility data attached to every listing. A farmer who plants Tulsi under Shravana nakshatra on a Jupiter day lists their surplus cuttings through the same system that told them when to plant. The buyer receives not just a cutting but the full cosmological context of its cultivation: the field state at germination, the guild companions it was grown alongside, the astrobotanical class and dosha profile of the harvest. This is what a produce market looks like when it is built on a knowledge graph rather than a logistics platform. The provenance is not a marketing claim. It is an immutable record in the attestation chain.

5.3 Federation Protocol and Node Transport

Each node — a van, a farm, a hackerspace, a gurukula, a research station, a traditional knowledge community — runs its own local-first Coherence Atlas instance. Shared canonical ontology (the stable shastra layer) provides interoperability across all nodes without imposing uniformity. Each node owns its sadhu overlay layer entirely: what it contributes to the federation, at what authority level, and with what access permissions is determined by the node, not by any central authority.

The transport layer between nodes is pull-on-use: when a Florida node needs species data and cultivation observations that its local datasets don't hold, it pulls from a Kerala node whose practitioners have documented those plants across decades of nakshatra-grounded observation. That data arrives with its source node, lineage, and authority level intact — it is Kerala's sadhu-layer knowledge, marked as such, enriching the Florida node's planting calendar with specificity that no generic database provides.

Resilience is the natural consequence of genuine exchange rather than a designed backup strategy. Data that has been pulled and used is cached locally. If the Kerala node goes offline — flood, power failure, infrastructure loss — the knowledge it contributed to the federation survives in every node that pulled from it. The archive is distributed because the exchange was real. The commu-

nity in Kerala that built that knowledge over generations retains full attribution in every node that holds it. Their contribution is immutable in the attestation chain regardless of what happens to their physical infrastructure.

Three principles govern the federation: canonical portability (the stable ontology moves between nodes without data loss or authority degradation), sadhu sovereignty (each node owns its overlay layer and controls its sharing), and guru transparency (AI inference is always marked as such and never promoted to canon without explicit confirmation from a node with shastra-level authority). No node can unilaterally alter another node's canonical layer. Proposals from guru-layer synthesis are broadcast as proposals and confirmed or rejected by the receiving node's own authority process.

5.4 Coherence Spectrum

Atlas is the technical substrate for three creative projects sharing the same ontology: *Coherence Spectrum* (persistent procedural civilization simulator in which Vedic frameworks function as gameplay mechanics; coherence spectrum from 0% Kali Band to 100% Satya Band; dosha-based morphogenesis generates creature forms), *Emergent Satya Yuga* (worldbuilding framework for the Kali-to-Satya transition; NPUs as Sanskrit-grammar-based meaning-computation units), and *The Devi Field* (speculative eco-mythic novel; Anika Rao, Eli Mercer, Maya Alvarez; stabilization through entrainment rather than apocalypse). The game simulates. The novel narrates. Atlas measures. Coherence is relational density. The Satya Yuga is the state where the graph is maximally coherent.

6 The Invitation

6.1 For Computational Jyotish Communities

VedAstro, PyJHora, Maitreya, Saravali: Atlas has a live relational graph in which nakshatra position is not a chart datum but a field coordinate connected to plant, sound, deity, marma, vastu zone, raga, and passage simultaneously. What Atlas needs: rigorous cross-validation of panchanga computations against established ephemeris platforms, dasa calculation testing, ayanamsa edge cases. Ask: run /field against your preferred platform for 30 dates and document discrepancies. If your system has an API, let's discuss interoperability.

6.2 For Sanskrit and Digital Humanities Scholars

GRETIL contributors, Vedavaapi developers, WSC Computational Sanskrit: the 26,000-chunk passage corpus is structured for annotation, every chunk a node connected to entities and cosmological coordinates. What Atlas needs: corpus expansion (30+ major Ayurvedic granthas not yet in machine-readable form, Gandharva Veda, Grihya Sutras, Gaudiya literature beyond Narottama), NLP tools for Sanskrit compound analysis, federation protocol design with Vedavaapi. Ask: review the passage corpus schema and identify priority ingestion sources.

6.3 For Biodynamic and Permaculture Practitioners

The nakshatra-grounded observation framework is something the biodynamic tradition has been practicing qualitatively for decades without a computational substrate that holds all the layers simultaneously. The Thun sidereal calendar maps planting types to zodiacal constellations. Coherence Atlas maps them to nakshatras, adds the graha lord, element, guna, tithi quality, and devi resonance, scores every plant in the dataset against the live field state, and returns guild companions alongside the primary plant recommendation. This is the same practice carried further into its own relational logic.

The exchange frontend extends this directly into commerce. A biodynamic farm growing by nakshatra calendar can list seeds, cuttings, plant starts,

and produce through the Coherence Atlas exchange layer with full cultivation provenance attached: field state at germination, planting muhurta, biodynamic preparation used, guild companions grown alongside, astrobotanical class, dosha profile. Buyers who understand this language — and increasingly they exist — are not buying a product. They are buying a specific moment of the field’s expression. That is a different market with different pricing logic and a different relationship between grower and buyer.

What Atlas needs: field validation of nakshatra-plant correspondences from practitioners who are actually planting by these principles and recording results. Local species datasets for your ecoregion. Biodynamic preparation timing records. Guild plant performance observations across seasons. And critically — growers willing to pilot the exchange frontend and tell us what the interface needs to serve an actual farm market. *Ask:* run /bhumi/agriculture for one planting season. Keep field notes. Return them as sadhu-layer annotations. If you want to pilot the exchange layer, reach out directly.

6.4 For Gaudiya Vaishnava Communities

This system was built from within the sampradaya. The Goloka layer, the Narotama corpus, the acintya bheda abheda architecture: all require theological review by practitioners with genuine sampradaya standing. What Atlas needs: validation of the Vraja forest mappings, pada corpus expansion beyond Narotama (Rupa Gosvami, Vishvanatha Chakravarti, Bhaktivinoda Thakura’s songs), field observation from practitioners who can identify when cosmological scoring is resonant versus discordant. *Ask:* read the Goloka documentation. Tell us what the tradition requires that the system is not yet providing.

6.5 For Sri Vidya and Tantric Scholars

The Nitya Devi registry and the Samaya/Kaula overlay architecture need review from practitioners with lineage standing in these traditions. Atlas has built the infrastructure. The tradition needs to populate and validate it. What Atlas needs: missing vahana and directional attributes for three Devis, Kaula overlay datasets, review of the capability signature vectors against lineage-specific interpretations. *Ask:* review the Devi dataset schema and propose corrections or extensions.

6.6 For Hackerspaces, Makers, and Open Hardware Builders

The knowledge tokenization architecture in Coherence Atlas is directly isomorphic to problems the open hardware community has been trying to solve for years. An immutable attestation record encoding source, lineage, authority level, and contribution chain for a nakshatra-plant relation is the same data structure as a verified component datasheet with immutable provenance — the kind of thing that matters when you are trying to establish whether a part is genuine, whether a schematic has been modified, whether a replacement component is actually compatible. The graph's relation schema (subject | predicate | object | source | confidence | stability | authority | attestation) is a general-purpose knowledge provenance system that happens to be populated with traditional Vedic knowledge. The schema itself is applicable to any domain where the integrity of information matters and where attribution chains need to survive forks and derivatives.

What this means practically: a maker community that builds on the Coherence Atlas federation protocol gets immutable part compatibility data, BOM (bill of materials) integrity verification, replacement part query substitution against a relational graph, and schematic versioning with provenance — all using the same attestation infrastructure that the Vedic knowledge layer runs on. The hardware application is not a side use case. It is the same problem at a different layer of manifestation.

A van-mounted NUC running a cosmological field computer is also a genuinely interesting hardware story in its own right. Under \$300, local-first, offline-capable, Intel NPU-accelerated via OpenVINO, audio output through MOTU M2. This is a replicable pattern that needs to run on more than one device to become a protocol. What Atlas needs: deployment in different contexts and on different hardware (Pi 5, Orange Pi with NPU, solar-powered field node), mesh networking experiments, and most critically a federation protocol for node discovery and graph synchronization that the hardware community helps design. *Ask*: replicate the deployment. Document what breaks. Propose a node discovery protocol. Then tell us how the attestation schema maps onto your own provenance problems.

6.7 For Indigenous and Traditional Ecological Knowledge Communities

The federation architecture was designed with your situation specifically in mind. A community running its own Coherence Atlas node owns that node entirely — its sadhu overlay layer, its species observations, its seasonal knowledge, its practitioner annotations. Nothing leaves the node without the community's explicit decision to share it, at whatever authority level they determine appropriate. The cryptographic attestation chain means that if knowledge is shared, it is permanently attributed to its source. It cannot be scraped, repackaged, or extracted without the attribution chain following it.

What the federation offers in return: your knowledge, shared on your terms, enriches the broader network and survives in it. If your physical infrastructure is threatened — flood, fire, displacement — the knowledge your community contributed remains in every node that pulled from it, permanently attributed to you. The archive is the community. The node is the institution. What Atlas needs: traditional ecological knowledge communities willing to pilot the node sovereignty model and help define what genuine data sovereignty looks like in practice. *Ask:* reach out. We will build the node infrastructure to your specifications before asking for anything in return.

6.8 For Sound Healers and Music Therapists

The clinical evidence base for raga therapy is real and growing: peer-reviewed controlled trials documenting Bhimpalasi for BP and HRV modulation, Todi for blood pressure and respiratory rate reduction across a 30-day intervention, Miyan ki Todi and Malkauns for attention modulation measured by ERP. The Pitta Raga Module (Nagarajan and Varma) provides a fully specified clinical protocol — ragas, time windows, listening duration, instrument, volume range — verified against Ayurvedic theory and clinical outcomes.

What does not yet exist is a computational tool that integrates this evidence with the live cosmological field: which raga is indicated not just for a dosha profile but for this dosha profile at this hour under this nakshatra with this tithi quality. Coherence Atlas holds all of those layers simultaneously. The therapeutic raga returned by the system is not arbitrary — it is derived through the nakshatra-graha-raga graph traversal, cross-referenced with the clinical evidence layer, and weighted by the current field state. What Atlas needs: clinical practition-

ers willing to run the system alongside their existing practice and document where the cosmological recommendations align with or diverge from clinical judgment. *Ask:* contact us with your clinical context. We will build a therapeutic interface adequate to your practice.

6.9 How to Connect

The system runs at localhost:5000 on kanjira. The /field endpoint requires no authentication and will tell you exactly what cosmological moment you are reading this in.

The offering cycle is the onboarding protocol:

impetus	-> what brought you here
cosmic context	-> what field state are you arriving in
offering plate	-> what you are bringing
seed	-> how it connects to what already exists
research lattice	-> what questions it opens
construction	-> software, wiki, dataset, visualization
git push	-> committing to the record
next cycle	-> the impetus that emerges

If you are reading this document, you have already completed the first step.

7 Appendix A: Current System State

As of April 2026.

Working: 217 routes in `kernel.py`. 27 engines, zero import failures. 163 Python files across 15 subdirectories in `npu_engine/`. 110 CSV files across 33 knowledge domains. 7,208 entities, 8,466 relational edges. 26,000-chunk JSONL passage corpus with vector search. `/field` returning complete panchanga + sound + plant + devi + marma + dasa + natal on every request. Agriculture PWA (offline-capable). Talachakra Studio (tabla synthesis, MIDI output, panchanga integration). Vishvakarma (GIS vastu, satellite imagery, 9 zones). Lila Streams (land design, permaculture mandala, sacred sites). Bandhu figure (6 rendering layers, toroidal breath animation, doctrine of signatures). Ring engine (10 ring types). 219 wiki pages. All six domain routes 200.

Not yet working: Vocal SynthDef (1,154 lines, no audio path). SuperCollider to MOTU (blocked on PipeWire 1.2.6 JACK shim). 4D projection (code exists, not exposed). Deha app beyond Bandhu placeholder. Seeds pipeline (4 briefs, not feeding graph). Journal UI (backend exists, no HTML). Editor (spec'd, not built). Systemd services (scripts ready, not installed).

Dataset gaps: 30+ major Ayurvedic granthas. Grihya Sutra ritual calendar. Carnatic melakarta-to-nakshatra mapping. Vraja forest coordinates. Vahana/directional data for 3 Nitya Devis. Yoni cross-species dataset.

8 Appendix B: Core Schemas

Entity: `entity_id` | `entity_type` | `name` | `name_key` | `bija` | `theta` | `phi` | `s_layer` | `element` | `guna` | `dosha` | `stability` | `authority`

Relation: `subject` | `predicate` | `object` | `source` | `confidence` | `stability` | `authority` | `attestation`

Passage: `chunk_id` | `source_text` | `tradition` | `entity_refs` | `domain` | `authority` | `passage_text`

Herb: `herb_id` | `name_sanskrit` | `name_latin` | `ay_rasa` | `ay_virya` | `ay_vipaka` | `ay_dosha_effect` | `tcm_nature` | `tcm_flavor` | `west_primary_action`

| biodynamic_day_type | nakshatra_correspondence | astrobotanical_class |
convergence_score | attestation

Attestation levels: OBSERVED:PRIMARY_TEXT (1.0) | OBSERVED:TRADITIONAL (0.8)
| OBSERVED:PEER_REVIEWED (0.9) | OBSERVED:PFAF (0.7) | SYNTHESIS (0.4) |
INTERPRETATION (0.2)

9 Appendix C: The Offering Cycle

impetus -- what is drawing attention right now
cosmic context -- what field state is this arising in
offering plate -- what tasks and research directions are proposed
seed -- how this connects to existing graph structure
research lattice -- what questions this opens
construction -- software, wiki pages, datasets, visualizations
README expansion -- documenting what was built
git push -- committing to the record
next cycle -- the impetus that emerges from the work

10 Appendix D: Glossary

Acintya bheda abheda — inconceivable simultaneous difference and non-difference; the philosophical foundation of the Atlas graph architecture, from Sri Chaitanya Mahaprabhu’s siddhanta. *Sat-sandarbha* — Jiva Gosvami’s six philosophical treatises establishing Gaudiya Vaishnava epistemology; the structural model for Atlas’s knowledge architecture. *Nama-rupa* — name and form; the two axes of every graph entity. *Lila* — divine play; the ontological status of the system. *Dakshina marga* — right-hand path; agamic, structured approach; corresponds to the shastra authority layer. *Vama marga* — left-hand path; Kaula, transgressive approach; corresponds to the sadhu overlay layer. *Ashtakala* — eight eternal periods of Goloka lila; the S0 temporal mapping layer. *Panchanga* — five-limbed almanac (tithi, nakshatra, vara, yoga, karana); the computational root of field_state(). *Shruti* — microtonal interval; each carrying a named semantic quality from the Sangitaratnakara. *Nakshatra* — one of 27 sidereal lunar mansions; primary field coordinate and coherence-scoring variable. *Tithi* — lunar day (Sun-Moon angular separation / 12°); maps to Nitya Devi, modulates rasa. *Muhurta* — auspicious time window; scored for 24 action

types by `intention_engine.py`. *Dasha* — Vimshottari planetary period system; biographical temporal context. *Yantra* — geometric compression of cosmological relations; emergent toroidal interference pattern. *Yoni* — animal archetype per nakshatra; cross-species relational mode signature. *Sampradaya* — lineage of transmission; encoded as tradition weight in the authority overlay system. *Seva* — selfless service as offering; the exchange model of the federation.

11 Appendix E: Primary Sources

Gaudiya Vaishnava: Bhagavad Gita As It Is (A.C. Bhaktivedanta Swami Prabhupada trans. and commentary), Srimad Bhagavatam (Bhagavata Purana), Brahma-samhita, Chaitanya-charitamrita. Narottama das Thakura: Prarthana, Prema-bhakti-candrika (primary Bandhu corpus). Srila Rupa Gosvami: Bhakti-rasamrita-sindhu. Srila Jiva Gosvami: Sat-sandarbha. Srila Bhaktivinoda Thakura: Tattva-viveka, Jaiva Dharma. Srila Vishvanatha Chakravarti Thakura.

Jyotish: Brihat Parashara Hora Shastra, Jaimini Upadesha Sutras, Brihat Samhita (Varahamihira), Brihat Jataka (Varahamihira), Saravali (Kalyana Varma), Surya Siddhanta. Computational: Swiss Ephemeris with JPL DE431.

Cosmology: Rigveda (10.121, 1.25, 10.14), Atharvaveda (10.7 Skambha hymn), Vishnu Purana (H.H. Wilson trans.), Bhagavata Purana (3.11.18-19), Brahmanda Purana, Matsya Purana (ch. 273).

Music: Natya Shastra (Bharata, Manomohan Ghosh trans.), Sangitaratnakara (Sarngadeva, Kunhan Raja trans.), Brihaddeshi (Matanga). Clinical: Nagarajan-Varma Pitta Raga Module; Bhimpalasi BP/HRV RCT; Todi 30-day intervention; Miyan ki Todi/Malkauns/Puriya ERP study.

Ayurveda: Charaka Samhita, Sushruta Samhita, Bhavaprakasha Nighantu, Ash-tanga Hridayam. Open datasets: Amidha (CC BY 4.0, 700+ herbs), PFAF (8,504 species), ETCM (403 herbs).

Biodynamic: Steiner, R. Agriculture Course (GA 327). DOK trial (21-year controlled comparison, Switzerland).

Nakshatra traditions: Taittiriya Samhita (Keith trans.); ICAR Navagraha Vatika and Nakshatra Vatika.

Tantra and Sri Vidya: Nitya Devi dhyana descriptions; Samaya and Kaula interpretive traditions preserved as separate overlay layers.

Ritual: Shrauta Sutras, Grihya Sutras, Shatapatha Brahmana.